A16728W1

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B4: NUCLEAR AND PARTICLE PHYSICS

TRINITY TERM 2024

Monday, 3 June, 2:30 pm – 4:30 pm

Answer **two** *questions.*

Start the answer to each question in a **fresh book.**

The use of approved calculators is permitted.

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

1. (a) Explain the concept of a nuclear form factor $F(q)$ for scattering of a charged particle with a nucleus, and write down a formula for $F(q)$ in terms of the charge density of the nucleus, $\rho(r)$.

A nucleus is assumed to have a charge density given by $\rho(r) = \rho_0$ for $r < R$ and $\rho(r) = 0$ for $r > R$. Calculate the form factor $F(q)$ for this nucleus. Sketch $|F(q)|^2$ in the range $0 < qR < 6$ using a logarithmic scale for the y-axis. [11]

(b) Evaluate the limits $\lim_{q\to 0} F(q)$, and $\lim_{q\to\infty} F(q)$ and give brief physical explanations for these two limits. The experimentally determined $F(q)$ show less sharp minima than the ones predicted by this model. Give one possible explanation for the origin of this discrepancy. [7]

(c) Why would we need to modify the Rutherford scattering cross-section in order to calculate the cross-section for electron-nucleus scattering with an electron energy $E \gtrsim m_e c^2$, where m_e is the mass of the electron and c is the speed of light in vacuum? [2]

(d) Explain how in principle a measurement of the differential cross-section for electron-nucleus elastic scattering can be used to determine the charge distribution for a nucleus. If we want to measure a charge distribution for a nucleus with a radius $R = 3$ fm, estimate the minimum electron energy that would be required. [5]

2. The binding energy of a nucleus with Z protons and A nucleons is given by the Semi-Empirical Mass Formula (SEMF)

$$BE(Z, A) = a_V A + a_S A^{2/3} + a_C \frac{Z^2}{A^{1/3}} + a_A \frac{(A/2 - Z)^2}{A} \pm \delta.$$

(a) Explain what general features of the nuclear interaction between two nucleons are required to explain the constant density of nucleons in nuclei. Hence explain the origin of the terms in the binding energy proportional to A and $A^{2/3}$. Using the uncertainty principle, estimate the approximate value of the momentum-squared, p^2, for a nucleon confined to a volume of radius r. If the nuclear force was attractive and of the same form as the Coulomb force but a factor 100 stronger, use the above result to estimate the approximate inter-nucleon separation for a bound system of two nucleons. [9]

(b) Explain briefly the concept of an exchange force. Given that the radius of a nucleus scales as $r = r_0 A^{1/3}$ with $r_0 \approx 1.2$ fm, estimate the mass of an exchange particle that would be required for the exchange force to have approximately the correct range. How could the exchange force concept be used to explain the peaks in the differential cross section for proton-neutron (pn) forward scattering (i.e. scattering at a a centre of mass angle $\theta = 0°$) and backward scattering ($\theta = 180°$) shown in the figure 1 [Hint: Rutherford scattering only has a forward peak.]

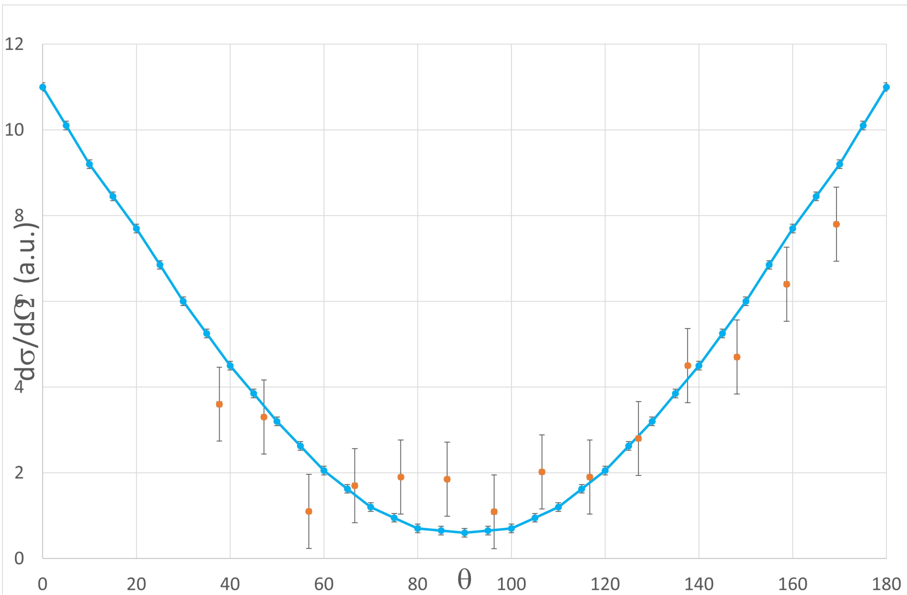

Figure 1: Differential cross section for proton-neutron scattering as a function of polar angle θ for a beam energy of 280 MeV. The points with error bars represent experimental data and the continuous curve is the result of a theoretical prediction.

[8]

(c) Consider nuclei with an odd-number of protons and an even-number of neutrons (o-e), and for a fixed value of A (with $A < 210$), determine the value of Z for which the isobar is stable. Which three processes are responsible for the decays of the unstable isobars? Explain what additional factor needs to be considered in examining the stability for even-even (e-e) and odd-odd (o-o) nuclei and how this can lead to more than one isobar that is stable with respect to the above three processes. [8]

3. The figure shows the meson nonet with $J^P = 0^-$.

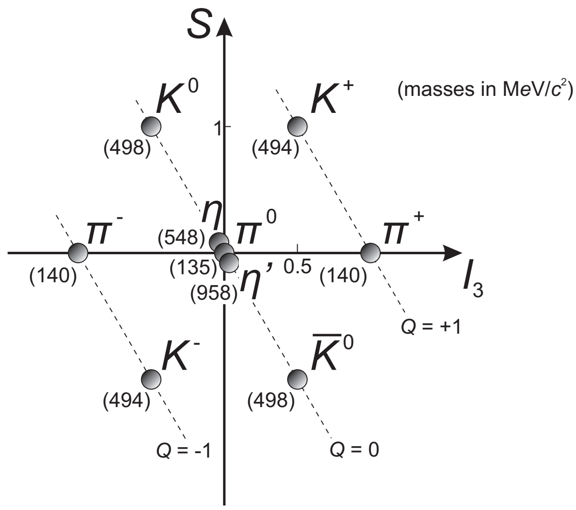

Figure 2: Mesons with $J^P = 0^-$.

(a) What are the two axes in this representation? What is their interpretation in the quark model? [4]

(b) Explain how in the quark model the spin and the parity of these particles can be derived from the properties of constituent quarks. Show using the charges of the constituent quarks, that the charge of the meson can be derived from $Q = I_3 + S/2$. Why are there nine mesons in this multiplet? [6]

Why are none of the mesons in this nonet stable? [1]

(c) The π^0 consists of a quark and a corresponding anti-quark. Show by using energy and momentum conservation that a massive particle cannot decay into a single particle. The π^0 therefore decays dominantly into two photons. Draw a Feynman diagram for this decay.

The decays $\pi^0 \to \gamma e^+e^-$ and $\pi^0 \to e^+e^-e^+e^-$ do also occur, but with smaller branching ratios. Explain why, and estimate by what factors they will be suppressed.

The η meson ($\eta = (1/\sqrt{6})(u\overline{u} + d\overline{d} - 2s\overline{s})$) can also decay by $\eta \to \gamma\gamma$. Explain why the branching ratio for this decay mode is smaller for the η than the π^0. [7]

(d) The charged pions and kaons can also decay by annihilation, but only with a coupling to a charged boson. This boson is much heavier than the decaying mesons. How is this possible? The lifetime of these mesons is relatively long. Why is that? [4]

The K^0 can not decay by the same decay mode as the π^0. Why? Draw a possible Feynman diagram for the decay of this meson. [3]

4. (a) Draw Feynman diagrams for the decays of the muon and the tau lepton. Are decays with hadrons in the final state possible for either of them? [3]

Explain why you would expect the ratio of partial decay rates Γ to be:

$$\frac{\Gamma(\tau^- \to e^- + \nu + \bar{\nu})}{\Gamma(\mu^- \to e^- + \nu + \bar{\nu})} \simeq \left(\frac{m_\tau}{m_\mu}\right)^5 \left(\frac{g_\tau}{g_\mu}\right)^2,$$

where g_τ and g_μ are the coupling factors to the W for the muon and the tau, respectively. [4]

(b) Evaluate the ratio g_τ/g_μ using $m_\tau = 1777.0\,\mathrm{MeV}/c^2$, $\tau_\tau = 2.91 \times 10^{-13}\,\mathrm{s}$, $m_\mu = 105.66\,\mathrm{MeV}/c^2$, $\tau_\mu = 2.197 \times 10^{-6}\,\mathrm{s}$, and $\mathrm{BR}(\tau^- \to e^-\nu\bar{\nu}) = 17.8\%$. What conclusion can we draw from this result for the coupling strength g_w, with which the W couples to the different types of leptons? [4]

(c) Draw a Feynman diagram at the quark level for the decay of the neutron. Here the decay involves a coupling of a quark to the W with a coupling strength g_{ud}. Label the vertices in your diagram with the respective coupling strengths. Experimentally it is found that $g_{ud}/g_\mathrm{w} \simeq 0.974$. A similar decay is the decay of the Λ^0 (quark content uds). Draw a Feynman diagram for this decay. Here the coupling of the W to the relevant quark has a relative strength $g_{us}/g_\mathrm{w} \simeq 0.221$. [3]

(d) Explain how the concept of weak interaction eigenstates that couple to the W with the same strength as the leptons can be maintained. How do the weak interaction eigenstates relate to the observable mass eigenstates? How many quarks are required for this relation to work? From the given data, find the value for this parameter. [5]

(e) Assuming this concept of universality for two quark generations of the weak interactions mediated by W, determine the branching ratio for electron, muon and hadronic decay modes $\mathrm{BR}(\tau^- \to e^-\nu\bar{\nu})$, $\mathrm{BR}(\tau^- \to \mu^-\nu\bar{\nu})$ and $\mathrm{BR}(\tau^- \to \nu + \mathrm{hadrons})$, where hadrons refers to any allowed hadronic final state (the masses of the lightest charmed mesons are $m_{D^\pm} = 1869.6\,\mathrm{MeV}/c^2$ and $m_{D^0} = 1864.8\,\mathrm{MeV}/c^2$). [4]

What consequence does the quark mixing have for the stability of heavy quarks, and the prevalence of light quarks in the universe? [2]